1. Matter Around Us

Chemistry | Standard 8 • Sheet 1/10 • FOUNDATION • Basic vocabulary

What is matter?

Matter is anything that has mass and occupies space. Objects around us are made of matter, although energy su

Materials

Materials have properties such as colour, hardness, solubility, and conductivity. These properties help us describe

Facts to Remember

• Matter has mass and occupies space.

• Materials can be described by their physical properties.

• Solubility describes how well a substance dissolves in a given solvent.

Check Your Understanding

1. Define matter.

2. Name three properties used to describe materials.

3. Is light matter? Give a reason.

2. States of Matter

Chemistry | Standard 8 • Sheet 2/10 • FOUNDATION • Compare states

Solids and liquids

A solid usually keeps its shape and volume. A liquid keeps its volume but takes the shape of its container.

Gases

A gas spreads to fill its container and can be compressed much more easily than a solid or liquid.

Facts to Remember

• A solid generally has a fixed shape and volume.

• A liquid has a fixed volume but takes its container's shape.

• A gas spreads to fill its container.

Check Your Understanding

1. Compare the shape of a solid and a liquid.

2. Why can a gas be compressed more easily?

3. Give one example of each state.

3. Particle Model

Chemistry | Standard 8 • Sheet 3/10 • FOUNDATION • Explain observations

Particles

Matter is made of tiny particles. In solids, particles are closely packed and vibrate about fixed positions; in liquids,

Temperature

Heating generally increases particle motion. This helps explain expansion and changes of state.

Facts to Remember

• Matter consists of particles too small to see individually.
• Particles in a solid vibrate around fixed positions.
• Heating generally increases particle motion.

Check Your Understanding

1. Use particles to explain why a liquid flows.
2. How does particle spacing differ in a gas and a solid?
3. What usually happens to particle motion when matter is heated?

4. Changes of State

Chemistry | Standard 8 • Sheet 4/10 • DEVELOPING • Process vocabulary

Heating

Melting changes a solid into a liquid. Evaporation changes liquid into vapour from its surface; boiling occurs throu

Cooling

Condensation changes gas into liquid. Freezing changes liquid into solid. Sublimation is a change from solid direc

Facts to Remember

• Melting is solid to liquid.
• Condensation is gas to liquid.
• Sublimation is solid directly to gas.

Check Your Understanding

1. Name the change from liquid to solid.
2. Distinguish evaporation from boiling.
3. What change occurs when water vapour forms droplets?

5. Pure Substances and Mixtures

Chemistry | Standard 8 • Sheet 5/10 • DEVELOPING • Classify materials

Pure substances

An element contains one kind of atom. A compound contains elements chemically combined in a fixed proportion.

Mixtures

A mixture contains two or more substances physically combined. Its components can often be separated by physi

Facts to Remember

- An element contains one kind of atom.
- A compound has elements chemically combined in a fixed proportion.
- A mixture contains substances physically combined.

Check Your Understanding

1. Classify oxygen as an element, compound, or mixture.
2. How does a mixture differ from a compound?
3. Why can mixture composition vary?

6. Separating Mixtures

Chemistry | Standard 8 • Sheet 6/10 • DEVELOPING • Choose a method

Physical methods

Filtration separates an insoluble solid from a liquid. Evaporation can recover a dissolved solid. Decantation separa

Other methods

A magnet can remove suitable magnetic materials. Distillation uses evaporation and condensation to separate su

Facts to Remember

• Filtration can separate an insoluble solid from a liquid.
• Evaporation can leave a dissolved solid behind.
• Distillation involves evaporation followed by condensation.

Check Your Understanding

1. Which method can separate sand from water?
2. How could salt be recovered from salt water?
3. Why would a magnet not separate dissolved salt from water?

7. Atoms, Molecules, and Symbols

Chemistry | Standard 8 • Sheet 7/10 • DEVELOPING • Read chemical language

Atoms and molecules

Atoms are the basic units of elements. A molecule is a group of atoms bonded together. Some substances, such

Symbols and formulae

Chemical symbols are abbreviated names for elements. A formula uses symbols and numbers to show which ato

Facts to Remember

• A chemical symbol represents an element.

• H2O contains hydrogen and oxygen in a 2:1 atom ratio.

• Subscripts in a formula show the number of atoms of the preceding element.

Check Your Understanding

1. What does the symbol O represent?

2. How many hydrogen atoms are shown in H2O?

3. What information does a chemical formula provide?

8. Physical and Chemical Changes

Chemistry | Standard 8 • Sheet 8/10 • CHALLENGE • Use evidence

Physical change

A physical change alters form or state without producing a new substance. Melting ice is physical: the substance

Chemical change

A chemical change produces one or more new substances. Evidence may include gas formation, a colour change

Facts to Remember

• A physical change does not create a new substance.

• A chemical change forms new substance(s).

• Observations such as bubbles or colour change can be clues, not conclusive proof alone.

Check Your Understanding

1. Why is melting ice a physical change?

2. Give two possible clues of a chemical reaction.

3. Why should a clue be interpreted with other evidence?

9. Chemical Reactions and Equations

Chemistry | Standard 8 • Sheet 9/10 • CHALLENGE • Interpret representations

Reaction

In a chemical reaction, atoms rearrange as reactants form products. A word equation names the reactants and pr

Conservation

In a balanced chemical equation, the number of atoms of each element is the same on both sides. Coefficients ch

Facts to Remember

• Reactants are the starting substances; products are formed.
• A balanced equation has equal counts of each element on both sides.
• Balancing changes coefficients, not chemical formula subscripts.

Check Your Understanding

1. What are reactants and products?
2. Why must atom counts match in a balanced equation?
3. Why is changing a subscript not an acceptable balancing step?

10. Acids, Bases, and Indicators

Chemistry | Standard 8 • Sheet 10/10 • CHALLENGE • Apply chemical properties

Acids and bases

Acidic and basic solutions have different chemical properties. Indicators change colour depending on the solution

pH and safety

The pH scale is commonly used to describe acidity or basicity: values below 7 are acidic, 7 is neutral, and above

Facts to Remember

• At typical classroom conditions, pH 7 is neutral.

• Values below 7 are acidic; values above 7 are basic.

• Indicators help identify acidic or basic solutions; unknown chemicals must not be tasted.

Check Your Understanding

1. What colour change does litmus commonly show in an acid?

2. Classify pH 3, pH 7, and pH 10.

3. Why is tasting a chemical an unsafe test?